TAMPINES MERIDIAN JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

CANDIDATE NAME	
CIVICS GROUP	

H2 GEOGRAPHY **9173/02**

Paper 2 **22 September 2025**
3 Hours

Additional Materials: 12 Page Answer Booklet
1 Insert

READ THESE INSTRUCTIONS FIRST

Write your answers in the answer booklet provided. You should follow the instructions on the front cover of the answer booklet. If you need additional answer paper, ask the invigilator for a continuation booklet.

Answer **all** questions.

The Insert contains all the Resources referred to in the questions.
You should make a reference to appropriate examples studied in the field or the classroom, even where the examples are not specifically requested by the question.
Diagrams and sketch maps should be drawn wherever they serve to illustrate an answer.
You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in the brackets [ ] at the end of each question or part question.

This document consists of **5** printed pages

Section A

Answer **both** questions

Cluster 4: Fieldwork

1 A group of ten 17-year-old students undertook an investigation to find out if the needs of the elderly in Singapore are being met. They identified Bishan, an estate located right in the middle of Singapore as their chosen neighbourhood for research. Bishan has a mix of public and private housing units. The public housing units range from 3-room flats to executive maisonettes. Similarly, the private housing comprises 4 condominiums and landed property units. Based on secondary research, the students found that the total population of elderly in the neighbourhood is 1298. 71% of the elderly population live in public housing and the remaining 29% live in private housing.

The following was their hypothesis:

"Elderly living in public housing have more unmet needs compared to elderly living in private housing."

The students conducted an on-site survey of the needs of elderly in both the public and private housing in the neighbourhood on four weekdays of their choice in the month of June. A total of 80 households with elderly residents were selected – 60 households chosen from 3-room flats in blocks 176, 177 and 178 of the public housing and 20 households from the private housing estate along Clover Avenue. As part of the safety precautions briefed by their Geography teacher, the students conducted the research in pairs and were explicitly forbidden to enter the homes of their respondents to conduct the survey.

Additionally, the students also observed the daily routine of two elderly men, who live in Block 175 of the public housing, for two hours on eight weekdays of their choice in the month of June. To improve the reliability of their investigation, they observed the men at different times during the day for the 8 weekdays and concluded that the old men spent most of the observed hours in the company of other elderly men chatting and playing board games at the Senior Citizens Corner at the void deck of Block 176.

Resource 1 shows the map of Bishan and the selected sites where the survey was conducted. Resources 2A and 2B show two sites where the survey was conducted. Resource 3 shows the questionnaire the students used for the on-site survey. Resource 4 shows the responses for selected parts of Question 6 of the questionnaire.

(a) Explain the strengths and weaknesses of the students' hypothesis. [4]

(b) With reference to Resources 1, 2A and 2B, explain the strengths and weaknesses of the student's choice of the neighbourhoods in which the elderly were surveyed. [6]

(c) Suggest how the students can address two issues related to research ethics that may arise in this geographical investigation. [4]

(d) Describe potential risks associated with this investigation at the sites shown in Resources 2A and 2B and suggest how the risks can be minimised. [5]

(e) The students concluded that the data collected using the questionnaire may not be accurate. With reference to Resource 3, explain how the questionnaire could be modified to improve the accuracy of the data collected. [6]

(f) Using Resource 4, sketch a suitable form of data representation to show the results for "accompany me to hospital". Justify your choice of method. [5]

(g) Using all resources available, evaluate the validity of the student's fieldwork. [10]

Cluster 3: Sustainable Future and Climate Change

2 Resource 5 shows the atmospheric carbon dioxide concentrations in parts per million from 800000 years ago till present.
Resource 6 shows the world's top eight greenhouse gas emitting countries in 2023.
Resource 7 shows the night time cityscape of Guangzhou, China's most liveable city, in 2024.

(a) Describe the trend in atmospheric carbon dioxide concentrations shown in Resource 5. [3]

(b) Explain how proxy indicators such as ice and ocean cores can provide evidence of past climate variability. [5]

(c) Account for the trend in carbon dioxide concentrations shown in Resource 5 after 1900. [6]

(d) With the use of suitable examples, explain why the term "urban liveability" is a subjective one. [4]

(e) Describe the global pattern of greenhouse gas emissions shown in Resource 6. [4]

(f) Using Resources 6 and 7, explain whether liveable cities like Guangzhou are also sustainable cities. [8]

Section B

Answer **either** question 3 **or** question 4.

Cluster 3: Sustainable Future and Climate Change

3 Evaluate the view that effective slum management is necessary for progress towards sustainable urban development. [20]

4 Evaluate the role firms play in responding to climate change. [20]

[End of Paper]